

Recent Advances and Updated Approaches in Pain Management (2023–2026)

Introduction

Pain management is an essential part of healthcare because pain can affect a patient's physical activity, sleep, emotional health, and overall quality of life. In previous years, opioids were commonly used for pain relief in many clinical conditions. However, the increasing concerns about opioid dependence and medication-related complications encouraged healthcare professionals to search for safer and more effective alternatives. Recent advances in pain management now focus on non-pharmacological therapies, multimodal analgesia, and personalized medicine. These updated approaches aim to improve patient outcomes while reducing the risks associated with long-term opioid use. This assignment reviews three recent peer-reviewed articles published between 2023 and 2026 that discuss new evidence and modern approaches in pain management.

Article 1 Summary

Full Citation

Moreau, S., Thérond, A., Cerda, I. H., Studer, K., Pan, A., Tharpe, J., Crowther, J. E., Abd-Elseyed, A., Gilligan, C., Tolba, R., Ashina, S., Schatman, M. E., Kaye, A. D., Yong, R. J., & Robinson, C. L. (2024). *Virtual Reality in Acute and Chronic Pain Medicine: An Updated Review*. *Current Pain and Headache Reports*, 28(9), 893–928. <https://doi.org/10.1007/s11916-024-01246-2>

Study Aim

The aim of this review article was to evaluate the recent evidence regarding the use of virtual reality (VR) in acute and chronic pain management.

Study Design / Methodology

This article was an updated review of recent studies, including randomized controlled trials and systematic reviews. The authors reviewed the use of VR in different pain conditions such as procedural pain, postoperative pain, and chronic musculoskeletal pain.

Main Findings

The review showed that VR therapy can reduce pain intensity and improve patient comfort during painful procedures and chronic pain treatment. VR was found to increase patient distraction and engagement, which may reduce the brain's focus on pain signals. The article also mentioned that more studies are still needed to evaluate long-term outcomes.

Clinical Relevance to Pain Management

This article is clinically important because it supports the use of VR as a non-pharmacological approach for pain relief. VR may help reduce the use of opioid medications and improve the patient experience during treatment.

Article 2 Summary

Full Citation

Yang, L., Lou, W., Jiang, Y., Yang, L., Wang, D., & Wang, J. (2025). *The clinical application progress of multimodal analgesia strategy in enhanced recovery after surgery: A narrative review*. *Frontiers in Pain Research*, 6, 1680157. <https://doi.org/10.3389/fpain.2025.1680157>

Study Aim

The purpose of this article was to discuss the recent progress of multimodal analgesia in Enhanced Recovery After Surgery (ERAS) programs and its role in postoperative pain management.

Study Design / Methodology

This article was a narrative review discussing current evidence related to multimodal analgesia strategies. The review included the use of non-opioid medications, regional anesthesia techniques, and combined pain management approaches used during perioperative care.

Main Findings

The article found that multimodal analgesia improves postoperative pain control while reducing opioid consumption. Patients receiving combined pain management methods showed faster recovery, earlier mobilization, and fewer opioid-related side effects.

Clinical Relevance to Pain Management

The findings support the use of multimodal analgesia as an important modern strategy in postoperative care. Combining different pain management methods may improve recovery and decrease complications associated with opioid medications.

Article 3 Summary

Full Citation

Ferreira do Couto, M. L., Fonseca, S., & Pozza, D. H. (2024). *Pharmacogenetic Approaches in Personalized Medicine for Postoperative Pain Management*. *Biomedicines*, 12(4), 729. <https://doi.org/10.3390/biomedicines12040729>

Study Aim

This article aimed to explain how pharmacogenetics can improve personalized pain management and influence patient responses to postoperative pain medications.

Study Design / Methodology

This review article discussed recent studies related to pharmacogenetics and postoperative pain control. The authors reviewed several important genes that may affect medication response, including CYP2D6, CYP2C9, OPRM1, and COMT.

Main Findings

The review showed that genetic differences between patients may affect the effectiveness and safety of pain medications. Some patients may experience poor pain relief, while others may have stronger side effects. The article suggested that genetic testing may improve medication selection and reduce opioid-related risks.

Clinical Relevance to Pain Management

This article highlights the importance of personalized medicine in pain management. Pharmacogenetic testing may help healthcare providers select safer and more effective medications based on each patient's genetic profile.

Conclusion

The three reviewed articles demonstrate that pain management is moving toward safer, more individualized, and evidence-based approaches. Virtual reality provides a useful non-drug option for pain relief, while multimodal analgesia improves postoperative recovery and reduces opioid use. Pharmacogenetic approaches may also help healthcare professionals choose medications that are more suitable for individual patients. These recent advances may improve patient safety, treatment outcomes, and overall quality of care in modern pain management practice.

References

Ferreira do Couto, M. L., Fonseca, S., & Pozza, D. H. (2024). *Pharmacogenetic Approaches in Personalized Medicine for Postoperative Pain Management*. *Biomedicines*, 12(4), 729. <https://doi.org/10.3390/biomedicines12040729>

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